

$\mathcal{O}$ -Slice Connectivity and Geometric Fixed Points

1 Input data: incomplete transfers and maximal admissible indices

Fix a finite group G . An N_∞ -operad (equivalently an indexing system) determines an *incomplete transfer system* $\mathcal{O}$, which we use only through the following consequence:

Definition 1 (Admissible orbits and maximal admissible index). *For each subgroup $H \leq G$, $\mathcal{O}$ specifies a collection of admissible transitive H -sets (equivalently orbits) H/K , written $K \rightarrow_{\mathcal{O}} H$. Define*

$$\mu_{\mathcal{O}}(H) := \max\{[H : K] \mid K \rightarrow_{\mathcal{O}} H\}.$$

Let $\mathcal{K}_{\max}(H)$ denote the (conjugation-stable) set of subgroups $K \leq H$ with $K \rightarrow_{\mathcal{O}} H$ and $[H : K] = \mu_{\mathcal{O}}(H)$.

For a finite H -set T , write $\mathbb{R}[T]$ for its real permutation representation, and for $K \leq H$ write

$$\rho_{H,K} := \mathbb{R}[H/K].$$

If H/K is transitive, then $(\rho_{H,K})^H \cong \mathbb{R}$ (constants), so $\dim(\rho_{H,K}^H) = 1$.

2 $\mathcal{O}$ -slice cells and the $\mathcal{O}$ -slice filtration

Work in a standard model of genuine G -spectra Sp^G (e.g. orthogonal G -spectra) or the equivalent presentable stable ∞ -category. We use standard facts: Sp^G is compactly generated; induction/restriction exist; and geometric fixed points exist.

Definition 2 ($\mathcal{O}$ -slice cells). *Fix $n \in \mathbb{Z}$. For each $H \leq G$ and each $K \in \mathcal{K}_{\max}(H)$ define the even and odd $\mathcal{O}$ -slice cells:*

$$C^{\mathrm{ev}}(H, K, k) := G_+ \wedge_H S^{k\rho_{H,K}}, \quad C^{\mathrm{odd}}(H, K, k) := G_+ \wedge_H S^{k\rho_{H,K}-1}, \quad (k \geq 0).$$

Define the $\mathcal{O}$ -slice dimension of these cells by

$$\dim_{\mathcal{O}}(C^{\mathrm{ev}}(H, K, k)) := k\mu_{\mathcal{O}}(H), \quad \dim_{\mathcal{O}}(C^{\mathrm{odd}}(H, K, k)) := k\mu_{\mathcal{O}}(H) - 1.$$

Let $\mathcal{C}_{\geq n}^{\mathcal{O}}$ be the set of $\mathcal{O}$ -slice cells with $\dim_{\mathcal{O}} \geq n$, and $\mathcal{C}_{< n}^{\mathcal{O}}$ those with $\dim_{\mathcal{O}} < n$.

Definition 3 ($\mathcal{O}$ -slice filtration). *Define*

$$\tau_{\geq n}^{\mathcal{O}} := \mathrm{Loc}(\mathcal{C}_{\geq n}^{\mathcal{O}}) \quad \text{and} \quad \tau_{< n}^{\mathcal{O}} := \mathrm{Loc}(\mathcal{C}_{< n}^{\mathcal{O}}),$$

the smallest localizing subcategories containing the indicated sets. A G -spectrum X is $\mathcal{O}$ -slice n -connective if $X \in \tau_{\geq n}^{\mathcal{O}}$.

Remark 1 (Connective G -spectra). *Throughout, connective means the underlying nonequivariant spectrum is connective. This suffices for the connectivity bounds below, since geometric fixed points preserve connectivity estimates in the way used here.*

3 Geometric fixed points: basic properties

Fix $H \leq G$. Let $\mathcal{P}_H$ be the family of proper subgroups of H . Let $\widetilde{E\mathcal{P}_H}$ be the cofiber of $E\mathcal{P}_H \rightarrow S^0$. Define the geometric fixed points functor by

$$\Phi^H(Z) := (\widetilde{E\mathcal{P}_H} \wedge Z)^H.$$

We use standard properties: Φ^H is exact and preserves all colimits; and for finite-dimensional H -representations V ,

$$\Phi^H(S^V) \simeq S^{V^H}.$$

Lemma 1 (Geometric fixed points of orbit representation spheres). *If H/K is transitive, then $(\rho_{H,K})^H \cong \mathbb{R}$ and hence*

$$\Phi^H(S^{k\rho_{H,K}}) \simeq S^k, \quad \Phi^H(S^{k\rho_{H,K}-1}) \simeq S^{k-1}.$$

Lemma 2 (Vanishing on induction from proper subgroups). *If $J < H$ and $Y \in \mathrm{Sp}^J$, then $\Phi^H(\mathrm{Ind}_J^H Y) \simeq 0$.*

Proof. In $\mathrm{Ind}_J^H Y \simeq H_+ \wedge_J Y$, all isotropy is subconjugate to J , hence belongs to $\mathcal{P}_H$. Smashing with $\widetilde{E\mathcal{P}_H}$ kills $\mathcal{P}_H$ -isotropy, so $(\widetilde{E\mathcal{P}_H} \wedge \mathrm{Ind}_J^H Y)^H \simeq *$. $\square$

4 A compactness and localization package

We use two standard facts about compact generation and localizations.

Lemma 3 (Compactness of $\mathcal{O}$ -slice cells). *Each $\mathcal{O}$ -slice cell $G_+ \wedge_H S^V$ with V finite-dimensional is compact in Sp^G .*

Proof. S^V is a finite H -CW spectrum, hence compact in Sp^H . Since G/H is finite, $G_+ \wedge_H (-)$ sends finite H -CW spectra to finite G -CW spectra, which are compact in Sp^G . $\square$

Proposition 1 (Slice localization triangle). *For each n , the inclusion $\tau_{\geq n}^{\mathcal{O}} \hookrightarrow \mathrm{Sp}^G$ admits a right adjoint $P_{\geq n}^{\mathcal{O}}$. Let $P_{< n}^{\mathcal{O}} X$ be the cofiber of the counit map $P_{\geq n}^{\mathcal{O}} X \rightarrow X$. Then there is a functorial cofiber sequence*

$$P_{\geq n}^{\mathcal{O}} X \longrightarrow X \longrightarrow P_{< n}^{\mathcal{O}} X$$

with $P_{\geq n}^{\mathcal{O}} X \in \tau_{\geq n}^{\mathcal{O}}$ and $P_{< n}^{\mathcal{O}} X \in \tau_{< n}^{\mathcal{O}}$. Moreover, if $P_{< n}^{\mathcal{O}} X \neq 0$, then there is some cell $C \in \mathcal{C}_{< n}^{\mathcal{O}}$ and a nonzero map $C \rightarrow X$.

Proof. Since $\tau_{\geq n}^{\mathcal{O}}$ is a localizing subcategory generated by a set of compact objects (Lemma 3), standard compactly-generated localization theory gives a right adjoint $P_{\geq n}^{\mathcal{O}}$. The cofiber $P_{< n}^{\mathcal{O}} X$ lies in the localizing subcategory generated by the complementary set of cells $\mathcal{C}_{< n}^{\mathcal{O}}$ (standard for cellular filtrations of this type), hence $P_{< n}^{\mathcal{O}} X \in \tau_{< n}^{\mathcal{O}}$. If $P_{< n}^{\mathcal{O}} X \neq 0$, then because $\tau_{< n}^{\mathcal{O}}$ is generated by compact cells, some compact generator maps nontrivially into $P_{< n}^{\mathcal{O}} X$, hence into X . $\square$

5 Mackey splitting and a detection lemma

Lemma 4 (Mackey restriction formula). *For $A \in \mathrm{Sp}^H$ there is a natural equivalence*

$$\mathrm{Res}_H^G(\mathrm{Ind}_H^G A) \simeq \bigvee_{[g] \in H \backslash G/H} \mathrm{Ind}_{H \cap gHg^{-1}}^H {}^g A,$$

with the $g = 1$ summand equal to A .

Lemma 5 (Detection from a nonzero cell map). *Let $C = C^{\mathrm{ev}}(H, K, k)$ (resp. $C^{\mathrm{odd}}(H, K, k)$) with $K \in \mathcal{K}_{\max}(H)$ and let $f : C \rightarrow X$ be nonzero. Choose such a nonzero map from cells in $\mathcal{C}_{<n}^{\mathcal{O}}$ with $|H|$ minimal. Then $\pi_k(\Phi^H(X)) \neq 0$ (resp. $\pi_{k-1}(\Phi^H(X)) \neq 0$).*

Proof. Write $C = \mathrm{Ind}_H^G S^W$ where $W = k\rho_{H,K}$ (even) or $W = k\rho_{H,K} - 1$ (odd). Restrict f to H and apply Lemma 4:

$$\mathrm{Res}_H^G(C) \simeq S^W \vee \bigvee_i \mathrm{Ind}_{J_i}^H B_i,$$

where each J_i is a proper subgroup of H . If the component $S^W \rightarrow \mathrm{Res}_H^G X$ were zero, then $\mathrm{Res}_H^G(f)$ factors through $\bigvee_i \mathrm{Ind}_{J_i}^H B_i$. Adjunction would then produce a nonzero map from a cell with subgroup $J_i < H$ into X , contradicting minimality of $|H|$. Thus $S^W \rightarrow \mathrm{Res}_H^G X$ is nonzero.

Apply Φ^H . By Lemma 2, $\Phi^H(\mathrm{Ind}_{J_i}^H B_i) \simeq 0$ for each $J_i < H$, so only the S^W -summand can contribute. Hence $\Phi^H(S^W) \rightarrow \Phi^H(X)$ is nonzero. By Lemma 1, $\Phi^H(S^W) \simeq S^k$ (even) or S^{k-1} (odd), giving the claimed nonzero homotopy class. $\square$

6 Connectivity in spectra

Let $\mathrm{Sp}_{\geq m} \subset \mathrm{Sp}$ denote the full subcategory of m -connective spectra.

Lemma 6 (Localizing generation of connective spectra). *For each $m \in \mathbb{Z}$, the localizing subcategory of Sp generated by spheres $\{S^r \mid r \geq m\}$ is exactly $\mathrm{Sp}_{\geq m}$.*

7 Main theorem

Theorem 1 ($\mathcal{O}$ -slice connectivity detected by geometric fixed points). *Let $X \in \mathrm{Sp}^G$ be connective. Then for every integer n ,*

$$X \in \tau_{\geq n}^{\mathcal{O}} \iff \forall H \leq G, \Phi^H(X) \in \mathrm{Sp}_{\geq \lceil n/\mu_{\mathcal{O}}(H) \rceil}.$$

Equivalently, for every $H \leq G$ and every $t < \lceil n/\mu_{\mathcal{O}}(H) \rceil$, one has $\pi_t(\Phi^H(X)) = 0$.

Proof. ($\Rightarrow$) Assume $X \in \tau_{\geq n}^{\mathcal{O}} = \mathrm{Loc}(\mathcal{C}_{\geq n}^{\mathcal{O}})$. Fix $H \leq G$. Since Φ^H is exact and preserves colimits, it sends $\tau_{\geq n}^{\mathcal{O}}$ into the localizing subcategory of Sp generated by the set $\Phi^H(\mathcal{C}_{\geq n}^{\mathcal{O}})$.

Consider a generator $C \in \mathcal{C}_{\geq n}^{\mathcal{O}}$ of the form $C^{\mathrm{ev}}(H', K', k)$ or $C^{\mathrm{odd}}(H', K', k)$. If H does not occur as a subgroup of a conjugate of H' , then $\Phi^H(C) \simeq 0$. If $H = H'$, then by Lemma 1,

$$\Phi^H(C^{\mathrm{ev}}(H, K', k)) \simeq S^k, \quad \Phi^H(C^{\mathrm{odd}}(H, K', k)) \simeq S^{k-1}.$$

Since $C \in \mathcal{C}_{\geq n}^{\mathcal{O}}$, we have $k\mu_{\mathcal{O}}(H) \geq n$ in the even case, and $k\mu_{\mathcal{O}}(H) - 1 \geq n$ in the odd case, hence in either case the sphere degree is $\geq \lceil n/\mu_{\mathcal{O}}(H) \rceil$.

If H is a proper subgroup of a conjugate of H' , then $\Phi^H(C)$ is a wedge of spheres in degrees *at least* those appearing in the $H = H'$ case (because fixed vectors only increase when passing to smaller subgroups). In particular, every nonzero $\Phi^H(C)$ is a wedge of spheres of degree $\geq \lceil n/\mu_{\mathcal{O}}(H) \rceil$.

Therefore $\Phi^H(X)$ lies in the localizing subcategory of $\mathbf{Sp}$ generated by spheres in degrees $\geq \lceil n/\mu_{\mathcal{O}}(H) \rceil$, which by Lemma 6 is $\mathbf{Sp}_{\geq \lceil n/\mu_{\mathcal{O}}(H) \rceil}$. Hence $\Phi^H(X)$ is $\lceil n/\mu_{\mathcal{O}}(H) \rceil$ -connective.

($\Leftarrow$) Assume that for every $H \leq G$, $\Phi^H(X)$ is $\lceil n/\mu_{\mathcal{O}}(H) \rceil$ -connective. Suppose for contradiction that $X \notin \tau_{\geq n}^{\mathcal{O}}$. By Proposition 1, $P_{<n}^{\mathcal{O}}X \neq 0$ and there exists a cell $C \in \mathcal{C}_{<n}^{\mathcal{O}}$ and a nonzero map $f : C \rightarrow X$. Choose such an f with $|H|$ minimal among the subgroup parameters appearing in C . By Lemma 5, this yields a nonzero class in $\pi_t(\Phi^H(X))$ where $t = k$ (even) or $t = k - 1$ (odd).

Since $C \in \mathcal{C}_{<n}^{\mathcal{O}}$, we have $k\mu_{\mathcal{O}}(H) < n$ (even) or $k\mu_{\mathcal{O}}(H) - 1 < n$ (odd), so in either case

$$t < \left\lceil \frac{n}{\mu_{\mathcal{O}}(H)} \right\rceil.$$

This contradicts the assumed $\lceil n/\mu_{\mathcal{O}}(H) \rceil$ -connectivity of $\Phi^H(X)$. Hence $X \in \tau_{\geq n}^{\mathcal{O}}$. □